1 Computing similarity transformations of operators

1.1 Motivation

Similarity transformations $B = P^{-1}AP$ are ubiquitous whenever a Hamiltonian is brought to a simpler (often diagonal) form. Common examples include the Bogoliubov transformations of quadratic Hamiltonians, displacement/squeezing of bosonic modes, polaron (Lang-Firsov) transformations, Schrieffer-Wolff transformations, and passage to the interaction picture or a rotating frame are all instances of this construction. Here, we focus on similarity transformation of the form $B \mapsto e^{A}Be^{-A}$ with $A^{\dagger} = -A$ (generator anti-Hermitian). Hence, $U = e^{A}$ is unitary and the transformation preserves Hermiticity and spectra. Similarity transformations are also called equivalence or canoncial transformation and play a big role in group theory.

1.2 Method A: The Hadamard lemma

Hadamard Lemma.
For any two operators A, B,

$$e^{A}Be^{-A} = \sum_{n=0}^{\infty} \frac{1}{n!} \operatorname{ad}_A^n(B), \qquad \text{with} \tag{1}$$

$$\operatorname{ad}_A^n(B) \equiv [A, B]_n \equiv [A, [A, B]_{n-1}], \quad \operatorname{ad}_A^0(B) \equiv [A, B]. \tag{2}$$

This series is especially helpful whenever the nested commutator yields zero at for a finite n or if the nested commutators eventually reproduce a small, recognizable pattern.

Strategy 1: Reduce to elementary operators.

Utilize that for any A, X, Y,

$$e^{A}(XY)e^{-A} = \big(e^{A}Xe^{-A}\big)\big(e^{A}Ye^{-A}\big). \tag{3}$$

It thus suffices to compute $e^{A}\,a\,e^{-A}$ and $e^{A}\,a^{\dagger}e^{-A}$ once and any polynomial in these follows by substitution and normal-ordering, with no further Hadamard series needed.

Strategy 2: The dagger shortcut.

If $A^{\dagger} = -A$, $U = e^{A}$ is unitary with $U^{\dagger} = e^{-A} = U^{-1}$, so $e^{A}Be^{-A} = UBU^{\dagger}$. Therefore, taking the dagger simply yields

$$\big(e^{A}Be^{-A}\big)^{\dagger} = e^{A}B^{\dagger}e^{-A}. \tag{4}$$

Once $e^{A}ae^{-A}$ is known, $e^{A}a^{\dagger}e^{-A}$ follows for free by Hermitian conjugation, roughly halving the bookkeeping.

1.3 Method B: The generator flow method (not proof read since not used yet)

Define $X(\theta) \equiv e^{\theta A}\, X\, e^{-\theta A}$, $X(0) = X$. Differentiating,

$$\frac{dX(\theta)}{d\theta} = [A, X(\theta)] = \mathrm{ad}_A\big(X(\theta)\big). \tag{5}$$

This ordinary differential equation (ODE) is exactly equivalent to Method A – the Hadamard series is its formal Taylor solution around $\theta = 0$ – but is can be more tractable in practice.

Key fact: closure ⇒ finite-dimensional linear ODE.

If ad_A maps a finite set of operators $\{X_1, \ldots, X_n\}$ into linear combinations of themselves, the operator ODE collapses to a matrix ODE:

$$\frac{d}{d\theta}\boldsymbol{X}(\theta) = M\,\boldsymbol{X}(\theta), \qquad [A, X_j] = \sum_i M_{ij} X_i, \qquad \boldsymbol{X}(\theta) = e^{\theta M}\boldsymbol{X}(0), \tag{6}$$

reducing the problem to exponentiating an ordinary $n \times n$ matrix.

1.4 When to prefer which method

Both methods are mathematically identical. Method A is fast when the pattern is recognizable by eye; Method B is more systematic for anything beyond the simplest single-operator case, especially with mixed or multiple generators to track jointly.

1.5 Scope: when this machinery fails

The closure property requires A to be at most quadratic in a set of canonically (anti-) commuting operators. For a genuinely quartic operator such as $H_{\mathrm{int}} = U \sum_i (n_i - \frac{1}{2})(n_{i+1} - \frac{1}{2})$, $\mathrm{ad}_A(H_{\mathrm{int}})$ does not close on a finite set for generic one-body A (e.g. $A \propto J$ or $A \propto T$) – $[J, H_{\mathrm{int}}]$ is generically nonzero and does not reduce to a small closed set of operators. The Hadamard series then cannot be resummed in closed form; one is limited to a perturbative treatment in the strength of A.

1.6 References

- J.J. Sakurai & J. Napolitano, *Modern Quantum Mechanics.*
- C. Gerry & P. Knight, *Introductory Quantum Optics.*
- D.F. Walls & G.J. Milburn, *Quantum Optics.*
- J.-P. Blaizot & G. Ripka, *Quantum Theory of Finite Systems*, MIT Press.
- A. Perelomov, *Generalized Coherent States and Their Applications*, Springer.
- J.W. Negele & H. Orland, *Quantum Many-Particle Systems.*
- B.C. Hall, *Lie Groups, Lie Algebras, and Representations.*